

Adapted from Graham's lecture slides.

第一章：概论

函数，函数式程序设计，
历史回顾，Haskell的特点和例子
课程内容概览

函数

- In Haskell, a function is a mapping that takes one or more arguments and produces a single result.

```
double x = x + x
```

- Computation by function application

```
double 3
= { applying double }
  3 + 3
= { applying + }
  6
```

- Computation by function application

```
double (double 2)
=   { applying the inner double }
double (2 + 2)
=   { applying + }
double 4
=   { applying double }
4 + 4
=   { applying + }
8
```

- Computation by function application

```
double (double 2)
=   { applying the outer double }
double 2 + double 2
=   { applying the first double }
(2 + 2) + double 2
=   { applying the first + }
4 + double 2
=   { applying double }
4 + (2 + 2)
=   { applying the second + }
4 + 4
=   { applying + }
8
```

函数式程序设计

- Functional programming is style of programming in which the basic method of computation is the application of functions to arguments;
- A functional language is one that supports and encourages the functional style.

例子

Summing the integers 1 to 10 in Java:

```
int total = 0;  
for (int i = 1; i ≤ 10; i++)  
    total = total + i;
```

The computation method is variable assignment.



Summing the integers 1 to 10 in Haskell:

```
sum [] = 0  
sum (x:xs) = x + sum xs  
  
sum [1..10]
```

The computation method is function application.



历史回顾

1930s:



Alonzo Church develops the lambda calculus,
a simple but powerful theory of functions.

历史回顾

1950s:



John McCarthy develops Lisp, the first functional language, with some influences from the lambda calculus, but retaining variable assignments.



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历史回顾

1960s:



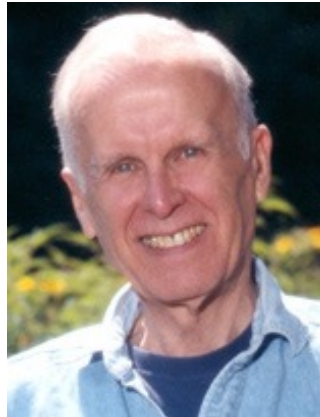
Peter Landin develops ISWIM, the first *pure* functional language, based strongly on the lambda calculus, with no assignments.



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历史回顾

1970s:



John Backus develops FP, a functional language that emphasizes *higher-order functions* and *reasoning about programs*.



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历史回顾

1970s:



Robin Milner and others develop ML, the first modern functional language, which introduced *type inference* and *polymorphic types*.



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历史回顾

1970s - 1980s:



David Turner develops a number of *lazy* functional languages, culminating in the Miranda system.



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历史回顾

1987:



An international committee starts the development of Haskell, a standard lazy functional language.



历史回顾

1990s:



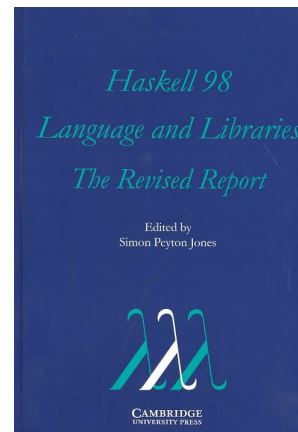
Phil Wadler and others develop *type classes* and *monads*, two of the main innovations of Haskell.



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历史回顾

2003:



The committee publishes the Haskell Report, defining a stable version of the language; an updated version was published in 2010.



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历史回顾

2010-date:



Standard distribution, library support, new language features, development tools, use in industry, influence on other languages, etc.



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Haskell的特点

- 简洁（声明式）：第2章，第4章
- 强有力的类型系统：第3章，第8章
- List comprehensions: 第5章
- 递归函数：第6章
- 高阶函数：第7章
- 表达副作用的函数：第10章，第12章
- Generic函数：第12章，第14章
- 惰性计算：第15章
- 程序推理：第16章，第17章

例1: 序列求和: 递归

```
sum []      = 0
sum (n:ns)  = n + sum ns
```

```
sum [1,2,3]
= { applying sum }
  1 + sum [2,3]
= { applying sum }
  1 + (2 + sum [3])
= { applying sum }
  1 + (2 + (3 + sum []))
= { applying sum }
  1 + (2 + (3 + 0))
= { applying + }
  6
```

例2: 快速排序: 递归

```
qsort []      = []  
qsort (x:xs) = qsort ys ++ [x] ++ qort zs  
              where  
                ys = [a | a ← xs, a ≤ x]  
                zs = [b | b ← xs, b > x]
```

例3: 序列求“和”一般化: 高阶函数

```
sum []          = 0
sum (n:ns)      = n + sum ns
```



```
foldr plus zero []          = zero
foldr plus zero (n:ns)      = plus n (foldr plus zero ns)

sum = foldr (+) 0
```

练习：用foldr定义（1）计算数字列表的乘积；（2）计算列表的长度。

例4: 生成无限序列: 惰性计算

```
cyclic = let x = 0 : y  
          y = 1 : x  
          in x
```

```
ns = 0 : foldr f 0 ns  
     where f n r = (1+n) : r
```

课堂内容一瞥: A Running Problem

如何构造出一个正确而且高效的解决最大子段和问题?

Maximum Segment Sum Problem

Given a list of numbers, find the maximum of sums of all *consecutive* sublists.

- $[-1, 3, 3, -4, -1, 4, 2, -1] \implies 7$
- $[-1, 3, 1, -4, -1, 4, 2, -1] \implies 6$
- $[-1, 3, 1, -4, -1, 1, 2, -1] \implies 4$



```
mss=0; s=0;
for(i=0;i<n;i++){
    s += x[i];
    if(s<0) s=0;
    if(mss<s) mss= s;
}
```



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课堂内容一瞥：A Running Problem

- 课程第一部分（Haskell）：利用高阶函数等的组合描述要做什么

```
mss = maximum . map sum . segs
where
  maximum = foldr max (-inf)
  sum = foldr (+) 0
  segs = concat . map tails . inits
  concat = ...
  tails = ...
  inits = ...
```



课堂内容一瞥：A Running Problem

- 课程第二部分 (Agda): 构建序列上的演算理论

```
map : ∀ {A B : Set} → (A → B) → List A → List B
map f [] = []
map f (x :: xs) = f x :: map f xs
```

```
map-compose-p : ∀ {A B C : Set} (f : A → B) (g : B → C) (x : List A) →
  (map g ∘ map f) x ≡ map (g ∘ f) x
map-compose-p f g [] =
  begin
  | (map g ∘ map f) []
  ≡()
  | map g (map f [])
  ≡()
  | map g []
  ≡()
  | []
  ≡()
  | map (g ∘ f) []
  ■
```

```
map-compose-p f g (x :: xs) =
  begin
  | (map g ∘ map f) (x :: xs)
  ≡()
  | map g (map f (x :: xs))
  ≡()
  | map g (f x :: map f xs)
  ≡()
  | g (f x) :: map g (map f xs)
  ≡( cong (g (f x) ::_) (map-compose-p f g xs) )
  | g (f x) :: map (g ∘ f) xs
  ≡()
  | (g ∘ f) x :: map (g ∘ f) xs
  ≡()
  | map (g ∘ f) (x :: xs)
  ■
```



课堂内容一瞥：A Running Problem

- 课程第二部分 (Agda): 构建序列上的演算理论

```
map-promotion :  
   $\forall \{A\ B : \text{Set}\} (f : A \rightarrow B)$   
   $\rightarrow \text{map } f \circ \text{flatten} \equiv \text{flatten} \circ \text{map } (\text{map } f)$   
  
reduce-promotion :  
   $\forall \{A : \text{Set}\} (\_ \oplus \_ : A \rightarrow A \rightarrow A) (e : A)$   
     $(p : \text{IsMonoid } \_ \oplus \_ e)$   
   $\rightarrow \text{reduce } \_ \oplus \_ e\ p \circ \text{flatten}$   
     $\equiv \text{reduce } \_ \oplus \_ e\ p \circ \text{map } (\text{reduce } \_ \oplus \_ e\ p)$ 
```



课堂内容一瞥: A Running Problem

- 课程第二部分 (Agda): 程序推理/程序演算
(+记号设计)

$$\begin{aligned} & mss \\ = & \{ \text{definition of } mss \} \\ & \uparrow / \cdot + / * \cdot segs \\ = & \{ \text{definition of } segs \} \\ & \uparrow / \cdot + / * \cdot ++ / \cdot tails * \cdot inits \\ = & \{ \text{map and reduce promotion} \} \\ & \uparrow / \cdot (\uparrow / \cdot + / * \cdot tails) * \cdot inits \\ = & \{ \text{Horner's rule with } a \odot b = (a + b) \uparrow 0 \} \\ & \uparrow / \cdot \odot \nearrow_0 * \cdot inits \\ = & \{ \text{accumulation lemma} \} \\ & \uparrow / \cdot \odot \nearrow_0 \end{aligned}$$



作业：

【1-1】 Define a function *product* that produces the product of a list of numbers, and show using your definition that $product [2,3,4] = 24$.

【1-2】 Define list length function using foldr.

【1-3】 How should the definition of the function qsort be modified so that it produces a reverse sorted version of a list?

请不要使用大语言模型！